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Math 31 — Differential Equations
Section 7.2.2

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38. $y' - y = 2\cos(5t) \mid y(0) = 0$

First, let us take the Laplace Transform of each side:

$$\mathcal{L}\{y' - y\} = \mathcal{L}\{2\cos(5t)\} \longrightarrow sY(s) - y(0) - Y(s) = \frac{2s}{s^2+25}$$

Factoring and adding over $y(0)$ we get:

$$Y(s)(s-1) = \frac{2s}{s^2+25} + 0$$

Dividing by $(s-1)$ results in:

$$Y(s) = \frac{2s}{(s-1)(s^2+25)}$$

Here we must take the inverse Laplace and thus need to make use of PFD. $(s-1)$ is a reducible linear term, however (s^2+25) does not have any real zeroes therefore we must make use of a linear term in our PFD.

$$\frac{2s}{(s-1)(s^2+25)} = \frac{As+B}{(s^2+25)} + \frac{C}{s-1}$$

Simplifying further:

$$\frac{2s}{(s-1)(s^2+25)} = \frac{As^2-As+Bs-B+Cs^2+25C}{(s-1)(s^2+25)}$$

Comparing coefficients we get:

$$A + C = 0$$
$$-A + B = 2$$
$$-B + 25C = 0$$

Setting up our augment matrix: $\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & -1 & 25 & 0 \end{array}\right) \longrightarrow$ RREF: $\left(\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{1}{13} \\ 0 & 1 & 0 & \frac{25}{13} \\ 0 & 0 & 1 & \frac{1}{13} \end{array}\right)$

Giving us:

$$A = -\frac{1}{13}, B = \frac{25}{13}, C = \frac{1}{13}$$

Rewriting our original fraction in its decomposed form gives us:

$$\frac{-\frac{1}{13}s+\frac{25}{13}}{(s^2+25)} + \frac{\frac{1}{13}}{(s-1)}$$

Simplyfing further we get:

$$Y(s) = \frac{25-s}{13(s^2+25)} + \frac{1}{13(s-1)}$$

To get the solution for our DE, we will now take the inverse Laplace:

$$y = \mathcal{L}^{-1}\left\{\frac{-s+25}{13(s^2+25)}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{13(s-1)}\right\}$$

Splitting the transformation results in:

$$y = -\frac{1}{13}\mathcal{L}^{-1}\left\{\frac{s}{s^2+25}\right\} + \frac{25}{13}\mathcal{L}^{-1}\left\{\frac{1}{s^2+25}\right\} + \frac{1}{13}\mathcal{L}^{-1}\left\{\frac{1}{(s-1)}\right\}$$

Getting the inverse Laplace transform into form:

$$y = -\frac{1}{13}\mathcal{L}^{-1}\left\{\frac{s}{s^2+25}\right\} + \frac{5}{13}\mathcal{L}^{-1}\left\{\frac{5}{s^2+25}\right\} + \frac{1}{13}\mathcal{L}^{-1}\left\{\frac{1}{(s-1)}\right\}$$

Finally:

$$\boldsymbol{\left[y = \frac{1}{13}e^t + \frac{5}{13}\sin(5t) - \frac{1}{13}\cos(5t)\right]}$$

39. $y'' + 5y' + 4y = 0 \mid y(0) = 1, y'(0) = 0$

Taking the Laplace of both sides gives us:

$$s^2Y(s) - sy(0) - y'(0) + 5(sY(s) - y(0)) + 4Y(s) = 0$$

Simplifying further:

$$s^2Y(s) - s + 5sY(s) - 5 + 4Y(s) = 0$$

Rearranging and factoring:

$$Y(s)(s^2+5s+4)=5+s$$

Isolating $Y(s)$:

$$Y(s)=\frac{5+s}{(s+1)(s+4)}$$

Completing PFD:

$$\frac{5+s}{(s+1)(s+4)}=\frac{A}{(s+1)}+\frac{B}{(s+4)}$$

Getting a common denominator:

$$\frac{5+s}{(s+1)(s+4)}=\frac{As+4A+Bs+B}{(s+1)(s+4)}$$

Comparing coefficients:

$$A+B=1$$

$$4A+B=5$$

Matrix of system is: $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ -4 & -1 & 5 \end{array}\right) \longrightarrow$ RREF $\left(\begin{array}{cc|c} 1 & 0 & \frac{4}{3} \\ 0 & 1 & -\frac{1}{3} \end{array}\right)$

Which gives us:

$$A=-\frac{4}{3}, B=-\frac{1}{3}$$

Thus our PFD result is:

$$\frac{4}{3(s+4)}-\frac{1}{3(s+1)}$$

Applying the inverse Laplace to both sides:

$$y=\mathcal{L}^{-1}\left\{\frac{4}{3(s+1)}\right\}-\mathcal{L}^{-1}\left\{\frac{1}{3(s+4)}\right\}$$

Which gives us our final solution of:

$$\boldsymbol{[y=\frac{4}{3}e^{-t}-\frac{1}{3}e^{-4t}]}$$

40. $y''-4y'=6e^{3t}-3e^{-t} \mid y(0)=1, y'(0)=-1$

Apply $\mathcal{L}\{\}$ to both sides:

$$s^2Y(s)-sy(0)-y'(0)-4sY(s)-4y(0)=\frac{6}{s-3}-\frac{3}{s+1}$$

Substituting our initial conditions:

$$s^2Y(s)-s+1-4sY(s)+4=\frac{6}{s-3}-\frac{3}{s+1}$$

Isolate $Y(s)$ and add over the -3 and $-s$:

$$Y(s)(s^2-4s)=\frac{6}{s-3}-\frac{3}{s+1}+s-5$$

Combining our first two fractions results in:

$$\frac{6s+6-3s+9}{(s-3)(s+1)}+s-5 \longrightarrow \frac{3s+15}{(s-3)(s+1)}+s-5$$

Adding the s and the -5:

$$\frac{3s+15+s^3-7s^2+7s+15}{(s-3)(s+1)} \longrightarrow \frac{s^3-7s^2+10s+30}{(s-3)(s+1)}$$

Dividing over the denominator from $Y(s)$:

$$\frac{s^3-7s^2+10s+30}{s(s-3)(s+1)(s-4)}$$

Setting up the PFD of our fraction:

$$\frac{s^3-7s^2+10s+30}{s(s-3)(s+1)(s-4)}=\frac{A}{s}+\frac{B}{s+1}+\frac{C}{s-4}+\frac{D}{s-3}$$

Adding each fraction:

$$\frac{As^3+Bs^3+Cs^3+Ds^3-6As^2-7Bs^2-2Cs^2-3Ds^2+5As+12Bs-3Cs-4Ds+12A}{s(s-3)(s+1)(s-4)}$$

The coefficient matrix for our numerator gives us:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ -6 & -7 & -2 & -3 & -7 \\ 5 & 12 & -3 & -4 & 10 \\ 12 & 0 & 0 & 0 & 30 \end{array}\right) \longrightarrow \text{RREF: } \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{5}{2} \\ 0 & 1 & 0 & 0 & -\frac{3}{5} \\ 0 & 0 & 1 & 0 & \frac{11}{10} \\ 0 & 0 & 0 & 1 & -2 \end{array}\right)$$

Thus our corresponding coefficients are:

$$A=\frac{5}{2}, B=-\frac{3}{5}, C=\frac{11}{10}, D=-2$$

Thus our decomposed fraction results in:

$Y(s) = \frac{5}{2s} - \frac{3}{5(s+1)} + \frac{11}{10(s-4)} - \frac{2}{(s-3)}$

Taking $\mathcal{L}^{-1}$ of both sides gives us:

$$\left[\boldsymbol{y = \frac{5}{2} - \frac{3}{5}e^{-t} + \frac{11}{10}e^{4t} - 2e^{3t}}\right]$$

41. $y'' + y = \sqrt{2}\sin\left(\sqrt{2}t\right) \mid y(0) = 10, y'(0) = 0$

$\mathcal{L}$ each side:

$s^2Y(s) - sy(0) - y'(0) + Y(s) = \frac{\sqrt{2}\sqrt{2}}{s^2+\left(\sqrt{2}\right)^2}$

Substituting $y(0)$ and $y'(0)$:

$s^2Y(s) - 10s + Y(s) = \frac{2}{s^2+2}$

Adding over the $10s$ and factoring $Y(s)$:

$Y(s)(s^2+1) = \frac{2}{s^2+2} + 10s$

Simplifying further:

$Y(s) = \frac{10s^3+20s+2}{(s^2+2)(s^2+1)}$

Performing PFD, Note: $(s^2+2)(s^2+1)$ are both irreducible:

$\frac{10s^3+20s+2}{(s^2+2)(s^2+1)} = \frac{As+B}{(s^2+1)} + \frac{Cs+D}{(s^2+2)}$

Combing the two fractions:

$\frac{As^3+2As+Bs^2+2B+Cs^3+Cs+Ds^2+D}{(s^2+1)(s^2+2)}$

Which gives us the following coefficent matrix:

$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 10 \\ 0 & 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 20 \\ 0 & 2 & 0 & 1 & 2 \end{array}\right) \longrightarrow \text{RREF: } \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right)$

Which gives us our decomposed fraction:

$\frac{10s+2}{(s^2+1)} - \frac{2}{(s^2+2)}$

Getting our fractions in form for $\mathcal{L}^{-1}$:

$Y(s) = \frac{10s}{s^2+1} + \frac{2}{(s^2+1)} - \frac{2}{s^2+2}$

Take the inverse Laplace:

$$\left[\boldsymbol{y = 10\cos(t) + 2\sin(t) - \sqrt{2}\sin\left(\sqrt{2}t\right)}\right]$$

42. $y'' + 9y = e^t \mid y(0) = 0, y'(0) = 0$

Setup $\mathcal{L}$, since y(0) = y'(0) = 0, I will skip the steps involving them:

$$s^2Y(s) + 9Y(s) = \frac{1}{(s-1)}$$

Factor LHS and islate $Y(s)$:

$Y(s) = \frac{1}{(s-1)(s^2+9)}$

(s^2+9) is non-reducible therefore our decomposed fraction will be of the following form:

$\frac{1}{(s-1)(s^2+9)} = \frac{As+B}{(s^2+9)} + \frac{C}{(s-1)}$

Adding both fractions:

$\frac{As^2-As+Bs-B+Cs^2+9C}{(s^2+9)(s-1)}$ Augmented matrix is:

$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 9 & 1 \end{array}\right) \longrightarrow \text{RREF: } \left(\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{1}{10} \\ 0 & 1 & 0 & -\frac{1}{10} \\ 0 & 0 & 1 & \frac{1}{10} \end{array}\right)$

Substituting and simplifying our fractions:

$\frac{-s-1}{10(s^2+9)} + \frac{1}{10(s-1)}$

Getting into form for inverse Laplace:

$$Y(s) = -\frac{s}{10(s^2+9)} - \frac{1}{10(s^2+9)} + \frac{1}{10(s-1)}$$

Which gives us our final answer of:

$$\boldsymbol{\left[y = -\frac{1}{10}\cos(3t) - \frac{1}{30}\sin(3t) + \frac{1}{10}e^t\right]}$$